2.46 Bivariate Data: PMCC Hypothesis Testing

Worked Solutions

1)

The significance level is the probability of rejecting the null hypothesis when in fact it is true.

2 a)

Either: Compare test statistic with critical value: $0.638 > 0.6215$ so Reject H_0 Or: Compare p-value with significance level: $0.0444 < 0.05$ so Reject H_0

2 b)

Either: Compare test statistic with critical value: $-0.456 > -0.4973$ so Fail to reject H_0 Or: Compare p-value with significance level: $0.0681 > 0.05$ so Fail to reject H_0

3 a)

$0.487 > 0.4575$ Reject H_0

3 b)

$0.628 < 0.6694$ Fail to reject H_0

3 c)

P$0.593 > 0.5155$ Reject H_0

3 d)

$0.213 > 0.1678$ Reject H_0

4 a)

$-0.453 < -0.4409$ Reject H_0

4 b)

$-0.829 < -0.6694$ Reject H_0

4 c)

$-0.306 > -0.4622$ Fail to reject H_0

4 d)

$-0.197 > -0.2070$ Fail to reject H_0

5)

Critical Region is [0.5751,1]

6)

Critical Region is $[-1, -0.5742]$

7 a)

$0.658 < 0.6664$ Fail to reject H_0

7 b)

$-0.567 > -0.6021$ Fail to reject H_0

7 c)

$0.621 < 0.6226$ Fail to reject H_0

7 d)

$-0.248 < -0.2353$ Reject H_0

8)

Critical Region is $[-1, -0.7545] \cup [0.7545,1]$

9)

Critical Region is $[-1, -0.2199] \cup [0.2199,1]$

10)

Let ρ be the population correlation coefficient between the amount of sugar consumed daily by adults and their blood glucose levels. H_0: $\rho = 0$ H_1: $\rho > 0$ $r_s = 0.382$ The critical value for $n = 28$ for a one-tailed test at a 5% significance level is 0.3172 $0.382 > 0.3172$ so we reject H_0 There is sufficient evidence to suggest that there is a positive correlation between the amount of sugar consumed daily by adults and their blood glucose levels.

11)

Let ρ be the population correlation coefficient between the number of hours spent on social media each day and self-reported feelings of loneliness in teenagers. H_0: $\rho = 0$ H_1: $\rho < 0$ $r_s = -0.297$ The critical value for $n = 70$ for a one-tailed test at a 5% significance level is 0.3060 $-0.297 > -0.3060$ so we fail to reject H_0 There is insufficient evidence to suggest that there is a negative correlation between the number of hours spent on social media each day and self-reported feelings of loneliness in teenagers.

12 a)

In the opening statement, ρ is the population corelation coefficient and not the sample correlation coefficient (that is r). r must be compared against -0.3665 and not 0.3665, as we are considering the bottom tail.

12 b)

Let ρ be the population correlation coefficient between the water temperature and the growth rate of the fish. H_0: $\rho = 0$ H_1: $\rho < 0$ $r = -0.364$ The critical value for $n = 40$ for a one-tailed test at a 1% significance level is 0.3665 $-0.364 > -0.3665$ so we fail to reject H_0 There is insufficient evidence to suggest that there is a negative correlation between the water temperature and the growth rate of the fish.

13)

Let ρ be the population correlation coefficient between the levels of air pollution and numbers of adults with respiratory illnesses. H_0: $\rho = 0$ H_1: $\rho \neq 0$ $r = 0.209$ The critical value for $n = 50$ for a two-tailed test at a 5% significance level is 0.2787 $0.209 < 0.2787$ so we fail to reject H_0 There is insufficient evidence to suggest that there is any correlation between the levels of air pollution and numbers of adults with respiratory illnesses.

14)

Let ρ be the population correlation coefficient between the daily intake of fruits and vegetables in adults and their BMI. H_0: $\rho = 0$ H_1: $\rho \neq 0$ $r = -0.301$ The critical value for $n = 100$ for a two-tailed test at a 1% significance level is 0.2565 $-0.301 < -0.2565$ so we reject H_0 There is sufficient evidence to suggest that there is some correlation between the daily intake of fruits and vegetables in adults and their BMI.

15 a)

The critical value for $n = 28$ for a two-tailed test at a 5% significance level is 0.3739, and not 0.3172. If the sample product moment correlation coefficient, r, is greater than the critical value, then we reject H_0, so Jodie has come to the wrong conclusion as $0.482 > 0.3172$ (and 0.482 is greater than the correct critical value of 0.3739, so the conclusion holds).

15 b)

Let ρ be the population correlation coefficient between the number of hours of sleep college students get each night and their stress levels. H_0: $\rho = 0$ H_1: $\rho \neq 0$ $r = 0.482$ The critical value for $n = 28$ for a two-tailed test at a 5% significance level is 0.3739 $0.482 > 0.3739$ so we reject H_0 There is sufficient evidence to suggest that there is some correlation between the number of hours of sleep college students get each night and their stress levels.